

Invertible Completions of Upper-Triangular Operator Matrices at Every Finite Matrix Size

Candidate full solution to the higher-dimensional question in arXiv:2202.04960

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Status. Candidate full solution, likely valid pending expert review. The criterion is necessary and sufficient for every finite matrix size and arbitrary Banach spaces, assuming only that the interior diagonal operators are inner regular.

1 Source question

Let $X_1, \dots, X_n$ be Banach spaces, let $D_i \in \mathcal{B}(X_i)$ be fixed, and let

$$T_n^d(A) = \begin{pmatrix} D_1 & A_{12} & \cdots & A_{1n} \\ 0 & D_2 & \cdots & A_{2n} \\ \vdots & \ddots & \ddots & \vdots \\ 0 & \cdots & 0 & D_n \end{pmatrix}, \quad A_{ij} \in \mathcal{B}(X_j, X_i), \quad i < j.$$

Sarajlija and Djordjević give a 3×3 completion theorem on arbitrary Banach spaces. In Section 3 of [1], PDF page 10, they ask whether the 2×2 and 3×3 results admit a natural generalization to all $n > 3$, anticipating multiple Fredholm-type assumptions and several left-invertible defect maps. The complete passage is reproduced below.

3 Extension to higher dimensions

It is a natural question whether [7, Theorem 2] and Corollary 2.2 admit a natural generalization from matrix dimensions $n = 2$ and $n = 3$ to higher matrix dimensions $n > 3$. Since Corollary 2.2 contains an additional apriori assumption (we demand that one of diagonal operators is Fredholm), one must suppose that a mentioned generalization would incorporate more than one apriori assumptions of the same type. Furthermore, regarding the form of Theorem 2.1, one must conclude that such a generalization would consider appearing more than two left invertible operators J_i similar to those from 1) c) in Theorem 2.1. Therefore, a logical conclusion is that such a generalization would be very technical and with cumbersome assumptions. This conclusion is further supported by some earlier works of the first author. Namely, statement [14, Theorem 2.13] confirms such a hypothesis.

The theorem below answers this question by isolating the exact recursive defect condition. It also sharpens the source's Theorem 2.1: its condition 1 is not only sufficient but necessary for an invertible 3×3 completion.

2 Proof intuition

An inner regular operator D_i splits into an isomorphism from a complement of $\mathcal{N}(D_i)$ onto $\mathcal{R}(D_i)$, together with a zero map from its kernel to its cokernel. After all invertible pieces are collected, ordinary Schur elimination reduces the entire completion problem to a single operator between defect sums:

$$\mathcal{N}(D_2) \oplus \cdots \oplus \mathcal{N}(D_n) \longrightarrow X_1/\mathcal{R}(D_1) \oplus \cdots \oplus X_{n-1}/\mathcal{R}(D_{n-1}).$$

The original zero pattern says exactly that this defect operator is upper triangular.

The remaining question is purely about Banach spaces: when does an invertible upper-triangular operator exist between two finite direct sums? Its first diagonal block must be left invertible and its last diagonal block must be right invertible. Split off those two matched pieces. Gaussian row and column operations leave the same problem with one fewer row and column. Iterating this operation gives a finite, exact recursive criterion. At $n = 3$ the last residual condition is precisely the quotient isomorphism appearing in the source paper.

3 The recursive defect relation

All direct sums below are finite topological direct sums; their particular standard norm is irrelevant up to isomorphism. For two lists of Banach spaces $(E_1, \dots, E_m)$ and $(F_1, \dots, F_m)$, write

$$\text{UT}(E_1, \dots, E_m; F_1, \dots, F_m)$$

if there is an invertible upper-triangular block operator

$$U : \bigoplus_{j=1}^m E_j \longrightarrow \bigoplus_{i=1}^m F_i, \quad U_{ij} = 0 \quad (i > j).$$

Lemma 1 (recursive triangular-isomorphism criterion). *For $m = 1$, $\text{UT}(E_1; F_1)$ holds exactly when $E_1 \cong F_1$. For $m \geq 2$, $\text{UT}(E_1, \dots, E_m; F_1, \dots, F_m)$ holds exactly when there exist left-invertible operators*

$$L : E_1 \rightarrow F_1, \quad J : F_m \rightarrow E_m$$

such that

$$\text{UT}(E_2, \dots, E_{m-1}, E_m/J(F_m); F_1/L(E_1), F_2, \dots, F_{m-1}) \quad (1)$$

holds. The ranges of L and J are automatically closed and complemented, so the displayed quotients are Banach spaces.

Proof. Suppose first that $U = (U_{ij})$ is invertible and upper triangular. Its first column is $(U_{11}, 0, \dots, 0)^t$. Since U is bounded below, $L := U_{11} : E_1 \rightarrow F_1$ is bounded below. More precisely,

$$F_1 \xrightarrow{\iota_1} \bigoplus_i F_i \xrightarrow{U^{-1}} \bigoplus_j E_j \xrightarrow{\pi_1} E_1$$

is a bounded left inverse of L . Thus L is left invertible. Surjectivity of the last row of U shows that $R := U_{mm} : E_m \rightarrow F_m$ is onto, and

$$F_m \xrightarrow{\iota_m} \bigoplus_i F_i \xrightarrow{U^{-1}} \bigoplus_j E_j \xrightarrow{\pi_m} E_m$$

is a bounded right inverse J of R . In particular, $J : F_m \rightarrow E_m$ is left invertible and

$$F_1 = L(E_1) \oplus F'_1, \quad E_m = E'_m \oplus J(F_m)$$

for closed complements F'_1 and E'_m .

Use a triangular domain operation to subtract from the E_1 coordinate the component that maps the remaining columns into $L(E_1)$. Use a triangular range operation to subtract from the earlier rows the components produced by $J(F_m)$, whose last-row image is all of F_m . Both operations are invertible. Relative to the displayed splittings they transform U into

$$\begin{pmatrix} L_0 & 0 & 0 \\ 0 & U' & 0 \\ 0 & 0 & R_0 \end{pmatrix},$$

where $L_0 : E_1 \rightarrow L(E_1)$ and $R_0 : J(F_m) \rightarrow F_m$ are isomorphisms, and

$$U' : E_2 \oplus \cdots \oplus E_{m-1} \oplus E'_m \longrightarrow F'_1 \oplus F_2 \oplus \cdots \oplus F_{m-1}$$

is still upper triangular. Since U is invertible, U' is invertible. Identifying the complements with the corresponding quotients proves (1).

Conversely, choose complements as above and let U' realize (1). Define U to be L_0 on E_1 , $R_0 = J^{-1}$ on $J(F_m)$, and U' on the remaining summands, with every other block zero. This operator is upper triangular in the original lists and is the direct sum of three isomorphisms. Hence it is invertible. $\square$

4 Defect reduction

Recall that $D \in \mathcal{B}(X)$ is inner regular precisely when its kernel and range are closed and complemented. In that case choose

$$X = \mathcal{N}(D) \oplus M, \quad X = \mathcal{R}(D) \oplus K.$$

Then $D|_M : M \rightarrow \mathcal{R}(D)$ is an isomorphism and $K \cong X/\mathcal{R}(D)$.

Lemma 2 (Schur reduction to the defects). *Assume $D_1, \dots, D_n$ are inner regular. There is a choice of the off-diagonal blocks A_{ij} for which $T_n^d(A)$ is invertible if and only if there is an invertible block operator*

$$S : \bigoplus_{j=1}^n \mathcal{N}(D_j) \longrightarrow \bigoplus_{i=1}^n X_i/\mathcal{R}(D_i), \quad S_{ij} = 0 \quad (i \geq j). \quad (2)$$

Moreover every strict upper-triangular S in (2) can be realized by a choice of the A_{ij} .

Proof. Choose complements M_i, K_i as above and identify K_i with $X_i/\mathcal{R}(D_i)$. Reorder the domain as $M \oplus N = (\bigoplus_i M_i) \oplus (\bigoplus_i \mathcal{N}(D_i))$ and the range as $R \oplus K = (\bigoplus_i \mathcal{R}(D_i)) \oplus (\bigoplus_i K_i)$. In these coordinates any completion has the form

$$\tilde{T} = \begin{pmatrix} P & Q \\ V & W \end{pmatrix} : M \oplus N \rightarrow R \oplus K.$$

The block $P : M \rightarrow R$ is upper triangular and its diagonal blocks are the isomorphisms $D_i|_{M_i}$. Hence P is invertible. Schur elimination gives

$$\begin{pmatrix} I & 0 \\ -VP^{-1} & I \end{pmatrix} \tilde{T} \begin{pmatrix} I & -P^{-1}Q \\ 0 & I \end{pmatrix} = \begin{pmatrix} P & 0 \\ 0 & W - VP^{-1}Q \end{pmatrix}.$$

Thus $\tilde{T}$ is invertible exactly when $S := W - VP^{-1}Q : N \rightarrow K$ is invertible.

The original block pattern permits a map from the j th input to the i th output only when $i \leq j$, while D_j vanishes on $\mathcal{N}(D_j)$. Every path contributing to $VP^{-1}Q$ has strictly increasing source-to-target index. Consequently $S_{ij} = 0$ whenever $i \geq j$.

Conversely, given such an S , define A_{ij} to equal the composition

$$X_j \longrightarrow \mathcal{N}(D_j) \xrightarrow{S_{ij}} K_i \hookrightarrow X_i$$

for $i < j$, using the bounded projection associated with the chosen domain splitting, and set all other components of A_{ij} to zero. Then $Q = V = 0$, $P = \bigoplus_i D_i|_{M_i}$, and the Schur complement is exactly S . $\square$

5 Full all-dimensional characterization

Theorem 3 (all- n invertible completion criterion). *Let $n \geq 2$, let $X_1, \dots, X_n$ be arbitrary Banach spaces, and let $D_i \in \mathcal{B}(X_i)$. Assume only that the interior operators $D_2, \dots, D_{n-1}$ are inner regular. Put*

$$E_r = \mathcal{N}(D_{r+1}), \quad F_r = X_r / \mathcal{R}(D_r), \quad 1 \leq r \leq n-1.$$

Then the following are equivalent:

1. *There are blocks $A_{ij} \in \mathcal{B}(X_j, X_i)$, $i < j$, for which $T_n^d(A)$ is invertible.*
2. *D_1 is left invertible, D_n is right invertible, and*

$$\text{UT}(E_1, \dots, E_{n-1}; F_1, \dots, F_{n-1})$$

holds, equivalently the finite recursion in Lemma 1 terminates in an isomorphism.

Proof. Suppose $T_n^d(A)$ is invertible. Its restriction to the first coordinate shows that D_1 is bounded below and has complemented range: a bounded left inverse is obtained from the first coordinate of $T_n^d(A)^{-1}$. Thus D_1 is left invertible. Applying $T_n^d(A)^{-1}$ to vectors supported in the last output coordinate gives a bounded right inverse for D_n . Hence D_n is right invertible. In particular both endpoint operators are inner regular, so Lemma 2 applies.

In (2), the first domain summand $\mathcal{N}(D_1)$ maps to zero and the last range summand $X_n / \mathcal{R}(D_n)$ receives only zero. Invertibility therefore forces both spaces to vanish, as already follows from endpoint one-sided invertibility. Deleting them turns (2) into an ordinary upper-triangular isomorphism

$$\bigoplus_{r=1}^{n-1} E_r \longrightarrow \bigoplus_{r=1}^{n-1} F_r.$$

This is exactly the displayed UT condition.

Conversely, endpoint one-sided invertibility and interior regularity make all D_i inner regular. A UT witness gives the strict defect isomorphism in (2) after adjoining the zero endpoint defect spaces. Lemma 2 realizes it by off-diagonal blocks and proves the resulting completion invertible. $\square$

Corollary 4 (3×3 strengthening). *Under the hypotheses of Theorem 2.1 of [1], an invertible completion exists if and only if D_1 is left invertible, D_3 is right invertible, and there are left-invertible maps*

$$J_1 : \mathcal{N}(D_2) \rightarrow X_1 / \mathcal{R}(D_1), \quad J_2 : X_2 / \mathcal{R}(D_2) \rightarrow \mathcal{N}(D_3)$$

such that

$$\frac{X_1/\mathcal{R}(D_1)}{J_1\mathcal{N}(D_2)} \cong \frac{\mathcal{N}(D_3)}{J_2(X_2/\mathcal{R}(D_2))}.$$

Thus condition 1 of that theorem is equivalent to its completion condition 2, not merely sufficient for it.

Proof. Apply Lemma 1 with $m = 2$ to Theorem 3. □

For $n = 2$, Theorem 3 reduces to the classical criterion $X_1/\mathcal{R}(D_1) \cong \mathcal{N}(D_2)$ of Han, Lee, and Lee [3]. For finite dimensional defect spaces, Lemma 1 becomes the usual sequence of prefix dimension inequalities together with equality of total dimensions.

6 Verification, scope, and novelty

The accompanying script constructs a nontrivial upper-triangular defect isomorphism with range-block dimensions $(3, 1, 1)$ and domain-block dimensions $(1, 1, 3)$, embeds it as the Schur complement of an invertible reduced block matrix, and checks a dimension pattern with an obstructed first recursive embedding. It also verifies the Schur determinant identity and strict zero pattern for random off-diagonal blocks. These checks are not part of the proof. (A genuinely nontrivial one-sided defect for a square diagonal block is inherently infinite-dimensional, so ordinary finite square matrices cannot model the whole completion theorem.)

The result is invariant under all choices of kernel/range complements: the statement uses canonical quotient spaces, and different complements are isomorphic to them. No separability, Hilbert structure, or Fredholm finite-dimensionality is used. The matrix size must be finite; infinite nests would require convergence and uniform-boundedness hypotheses absent from the source question.

A bounded novelty check on 17 August 2026 covered the run’s four lightweight indexes, the local full sources for arXiv:2108.12153, 2110.07387, 2112.02350, and 2202.04960, and exact-title/core-keyword web searches through that date. Theorem 2.13 of [2], cited by the source, gives a sufficient condition requiring adjacent defect spaces to be isomorphic and a weaker Hilbert-space necessity statement. The other located papers concern spectral stability or Fredholm/Weyl completion, rather than the exact ordinary-invertibility criterion above. No recursive necessary-and-sufficient Banach-space result, nor the 3×3 necessity upgrade in Corollary 4, was located. Novelty confidence is moderate pending a specialist search, especially in the block-operator monograph literature.

7 Human review recommendation

First check that the Schur complement in Lemma 2 retains the strict index pattern; this is the bridge from operator completion to defect matching. Then check the two Gaussian eliminations in Lemma 1, including preservation of upper triangularity. Finally compare Corollary 4 line by line with condition 1(c) of the source’s Theorem 2.1. Those checks settle both the all- n result and the claimed 3×3 strengthening.

References

- [1] N. Sarajlija and D. S. Djordjević, *Completion problem of upper triangular 3×3 operator matrices on arbitrary Banach spaces*, arXiv:2202.04960v4 (2025), published in Bull. Iranian Math. Soc. 51 (2025), Paper 85.

- [2] N. Sarajlija, *Perturbing the spectrum of operator $T_n^d(A)$* , arXiv:2108.12153; RACSAM 117 (2023), Paper 10, especially Theorem 2.13.
- [3] J. K. Han, H. Y. Lee, and W. Y. Lee, *Invertible completions of 2×2 upper triangular operator matrices*, Proc. Amer. Math. Soc. 128 (2000), 119–123.